Abelian Varieties

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Lecture 1

Recall that an *elliptic curve* E over a field k, which is a proper curve which is also a group. In particular, they are projective (in fact, we can write it as a nonsingular plane cubic), and that the group is commutative. If k is algebraically closed, then $E(k)$ is divisible, and for all $n \in \mathbb{N}$,

$$E[n] = \ker(n\cdot : E(k) \to E(k))$$

is finite. In particular, if n is invertible in k, then

$$E[n] = (\mathbb{Z}/n\mathbb{Z})^2$$

On the other hand, if $\mathrm{char}(k) = p > 0$, then either

$$E[p^r] = \mathbb{Z}/p^r\mathbb{Z}$$

for all $r \geq 1$, or

$$E[p^r] = 0$$

for all $r \geq 1$.

The *Abel-Jacobi* theorem gives a homomorphism

$$E(k) \to \mathrm{Cl}^0(E) = \frac{\{\text{degree 0 divisors}\}}{\{\text{principal divisors}\}}$$

Given by $P \mapsto (P) - (0)$. In fact, by Riemann-Roch, this is an isomorphism.

Abelian varieties are "higher dimensional analogues of an elliptic curve". That is, they are a proper variety over k which is a group. If X is an abelian variety, we will see:

- X is projective and nonsingular,
- X is commutative,
- if k is algebraically closed, then $X(k)$ is divisible, with

  $$X[n] \cong (\mathbb{Z}/n\mathbb{Z})^{2g}$$

  where $g = \dim(X)$, n invertible in k. If $\mathrm{char}(k) = p > 0$, then

  $$X[p^r] = (\mathbb{Z}/p^r\mathbb{Z})^t$$

  with $0 \leq t \leq g$.
- and an analogue of the Abel-Jacobi theorem, there exists another abelian variety $\widehat{X}$, with $\dim(\widehat{X}) = g$, called the dual abelian variety, and an isomorphism

  $$\widehat{X}(k) \cong \mathrm{Pic}^0(X) \subseteq \mathrm{Pic}(X)$$

  where $\mathrm{Pic}(X)$ is the group of isomorphism classes of line bundles on X, which is in turn, isomorphic to the divisor class group. Moreover, we have a morphism $X \to \widehat{X}$, which is surjective with finite kernel. Note that we don't need to have $X \cong \widehat{X}$.

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In contrast to the one-dimensional case, we need a lot of algebraic geometry, especially the cohomology of coherent sheaves to prove these. One reason is that unlike elliptic curves, we don't have nice equations for them.

Conventions for this course: Every ring will be commutative unital, and ring homomorphisms will be unital. If $f : A \to B$ is a ring homomorphism, we say B is an A-algebra. We'll write ab for $f(a)b$. An A-module is *finite* if it is finitely generated. An A-algebra B is *finite* if it is finite as an A-module, and it is of *finite type* if it is finitely generated as an A-algebra.

A *family* $(x_i)_{i\in I}$ of elements of a set S is just $x_i \in S$, indexed by a set I. This is the same as a function $I \to S$. We'll write $\#S$ for the size of a set S, $\subset$ and $\subseteq$ will be used interchangably. Finally,

$$\mathbb{N} = \{0, 1, \dots\}$$

1 Kähler differentials

Let $\varphi : A \to B$ be a ring homomorphism. We'll define a B-module $\Omega_{B/A}$, which is the module of (Kähler) differentials, with an additive map $\mathrm{d} = \mathrm{d}_{B/A} : B \to \Omega_{B/A}$, such that

$$\begin{aligned} \mathrm{d}(b_1 b_2) &= b_1 \mathrm{d}b_2 + b_2 \mathrm{d}b_1 \\ \mathrm{d}a = \mathrm{d}(\varphi(a)) &= 0 \text{ for all } a \in A \end{aligned}$$

In particular, d is an A-linear map.

Definition 1.1

We define $\Omega_{B/A} = \mathcal{P}/\mathcal{Q}$, where

$$\begin{aligned} \mathcal{P} &= \text{free } B\text{-module on symbols } [b] \text{ for } b \in B \\ \mathcal{Q} &= \text{submodule generated by } [a], [b_1 + b_2] - [b_1] - [b_2], [b_1 b_2] - b_1[b_2] - b_2[b_1] \end{aligned}$$

and

$$\mathrm{d}b = [b] \mod \mathcal{Q}$$

An *A-derivation of B into a B-module M* is an additive map $\mathrm{D} : B \to M$, such that

$$\begin{aligned} \mathrm{D}(A) &= 0 \\ \mathrm{D}(b_1 b_2) &= b_1 D(b_2) + b_2 D(b_1) \end{aligned}$$

We will write $\mathrm{Der}_A(B, M)$ for the set of A-derivations $B \to M$. This is a B-module, with $(b'\,\mathrm{D})(b) = b'\,\mathrm{D}(b)$. Moreover, $\mathrm{Der}_A(B, \cdot)$ is a functor. That is, if we have a B-module homomorphism $f : M \to M'$, then we have a map

$$\mathrm{Der}(A, M) \to \mathrm{Der}(A, M')$$

given by $\mathrm{D} \mapsto f \circ \mathrm{D}$. In particular, $(\Omega_{B/A}, \mathrm{d}_{B/A})$ is the *universal* derivation.

Proposition 1.2. Suppose M is a B-module, then there is an isomorphism of B-modules,

$$\begin{aligned} \mathrm{Hom}_B(\Omega_{B/A}, M) &\cong \mathrm{Der}_A(B, M) \\ \psi &\mapsto \psi \circ \mathrm{d}_{B/A} \end{aligned}$$

Proof. The fact that it is a B-module homomorphism is trivial. Let $\mathrm{D} \in \mathrm{Der}_A(B, M)$. Set $\widetilde{\psi} : \mathcal{P} \to M$ to be B-linear, with $\widetilde{\psi}([b]) = D(b)$. Since D is a derivation, $\widetilde{\psi}(Q) = 0$, so we have a map $\psi : \mathcal{P}/\mathcal{Q} = \Omega_{B/A} \to M$, with $\psi \circ \mathrm{d} = D$.

On the other hand, if $\psi \circ \mathrm{d} = 0$, then $\psi = 0$ since $\Omega_{B/A}$ is generated by the image of d. □

Lecture 2

Another characterisation of $\Omega_{B/A}$ is as follows. Consider the map $\mu : B \otimes_A B \to B$, given by $\mu(b_1 \otimes b_2) = b_1 b_2$. In particular, μ is a B-algebra homomorphism, for each of the B-algebra structues on $B \otimes_A B$. Let $J = \ker(\mu)$.

Then J/J^2 is a $B \otimes_A B$-module, killed by J, so it is a B-module. In particular, all of the B-module structures on it agree.

Let

$$\begin{aligned} \mathrm{d}' : B &\to J/J^2 \\ b &\mapsto (1 \otimes b - b \otimes 1) \mod J^2 \end{aligned}$$

Then d' is an A-derivation. To see this,

$$\mathrm{d}'(a) = 1 \otimes a - a \otimes 1 = 0$$

for all $a \in A$. Next,

$$\begin{aligned} b_1 \mathrm{d}'(b_2) &= (b_1 \otimes 1)(1 \otimes b_2 - b_2 \otimes 1) \mod J^2 \\ &= b_1 \otimes b_2 - b_1 b_2 \otimes 1 \mod J^2 \end{aligned}$$

and by symmetry,

$$b_2 \mathrm{d}'(b_1) = 1 \otimes b_1 b_2 - b_1 \otimes b_2 \mod J^2$$

Adding these together gives the Leibniz rule. By the universal property, there exists a unique B-linear map $\psi : \Omega_{B/A} \to J/J^2$, such that $\psi \circ \mathrm{d} = \mathrm{d}'$.

Proposition 1.3. ψ is an isomorphism.

So we could have defined $(\Omega_{B/A}, \mathrm{d})$ as $(J/J^2, \mathrm{d}')$ instead.

Proof. Consider the map

$$\begin{aligned} \varphi : B \otimes_A B &\to \Omega_{B/A} \\ \varphi(b_1 \otimes b_2) &= b_1 \, \mathrm{d}(b_2) \end{aligned}$$

Now

$$J = \left\{ \sum b_i \otimes b_i' \;\middle|\; \sum b_i b_i' = 0 \right\}$$

We claim that J is generated (as a B-module) by elements of the form $1 \otimes b - b \otimes 1$. To see this,

$$\sum b_i \otimes b_i' = \sum (b_i \otimes 1)(1 \otimes b_i' - b_i' \otimes 1) + \sum b_i b_i' \otimes 1$$

and the last term is zero. So it is a B-module under the $b \otimes 1$ action. Now

$$\varphi((1 \otimes b - b \otimes 1)(1 \otimes b' - b' \otimes 1)) = 0$$

and so φ vanishes on J^2, and it factors through $(B \otimes_A B)/J^2$. It remains to check that $\varphi \circ \psi = \mathrm{id}$ and $\psi \circ \varphi|_{J/J^2} = \mathrm{id}$. We leave this as an exercise. □

Remark 1.4. Let $C = (B \otimes_A B)/J^2$, then we have

$$\begin{array}{ccccc} B & \xrightarrow{\cong} & B \otimes_A A & \hookrightarrow & C \\ & \searrow^{\mathrm{id}} & & & \downarrow{\scriptstyle \mu} \\ & & & & B \end{array}$$

This gives us an isomorphism of $B = B \otimes_A A$-algebras,

$$B \oplus J/J^2 \cong C$$

The ring structure on the left hand side is given by

$$(b, f) \cdot (b', f') = (bb', bf' + b'f)$$

From the definition, any commutative square of rings

$$\begin{array}{ccc} B & \longrightarrow & B' \\ \uparrow & & \uparrow \\ A & \longrightarrow & A' \end{array}$$

induces a map

$$\Omega_{B|A} \to \Omega_{B'|A'}$$

which is B-linear, commuting with d. Moreover, this is transitive for $B \to B' \to B''$.

Proposition 1.5. (i) If $B' = B \otimes_A A'$, then

$$\Omega_{B|A} \otimes_A A' \cong \Omega_{B|A} \otimes_B B' \cong \Omega_{B'|A'}$$

(ii) if $A = A'$, and $B' = S^{-1}B$ is a localisation of B, then

$$\Omega_{S^{-1}B|A} = S^{-1}\Omega_{B|A}$$

Proof. Exercise. □

Example 1.6

If $B = A[t_1, \ldots, t_n]$ is the polynomial algebra, then

$$\Omega_{B|A} = \oplus_{i=1}^{n} B\mathrm{d}t_i$$

is free. To see this,

$$B \otimes_A B = A[\{t_i \otimes 1, 1 \otimes t_j\}] = A[\{t_i \otimes 1, z_i\}] = B[\{z_i\}]$$

where $z_i = 1 \otimes t_i - t_i \otimes 1$. Using this, the map $\mu : B \otimes_A B \to B$ is just $z_i \mapsto 0$ and the identity map on B. So we have that

$$J = \langle z_1, \ldots, z_n \rangle$$

and

$$J^2 = \langle z_i z_j \rangle$$

Thus, J/J^2 is the free module

$$\bigoplus_i B \cdot (z_i \mod J^2)$$

But $z_i \mod J^2$ is just $\mathrm{d}'(z_i) = \mathrm{d}t_i$.

We have two exact sequences for Ω:

Proposition 1.7 (transitivity). if $A \to B \to C$ are ring homomorphisms, then

$$\Omega_{B|A} \otimes_B C \longrightarrow \Omega_{C|A} \longrightarrow \Omega_{C|B} \longrightarrow 0$$

is exact.

Proposition 1.8 (second exact sequence). If $A \to B \twoheadrightarrow C$ are ring homomorphisms, with say $C = B/I$, then

$$I/I^2 \longrightarrow \Omega_{B|A} \otimes C \longrightarrow \Omega_{C|A} \longrightarrow 0$$

is exact, where $f \mod I^2 \mapsto \mathrm{d}f \otimes 1$. This is an exact sequence of C-modules.

Proof of both have been left as exercises.

Corollary 1.9. If $B = A[\{x_i\}]$, and $C = B/\langle\{f_j\}\rangle$, then

$$\Omega_{C/A} = \frac{\bigoplus_i C\,\mathrm{d}x_i}{\sum C\,\mathrm{d}f_j}$$

In this case,

$$\mathrm{d}f_j = \sum_i \frac{\partial f_j}{\partial x_i}\mathrm{d}x_i$$

1.1 Sheafification

Let $f : X \to Y$ be a morphism of schemes. We would like to define a quasicoherent $\mathcal{O}_X$-module $\Omega_{X/Y}$, and a map

$$\mathrm{d} : \mathcal{O}_X \to \Omega_{X/Y}$$

which is $f^{-1}\mathcal{O}_Y$-linear, and satisfies the Leibniz rule for sections. This is called the *sheaf of relative differentials*, or the *cotangent sheaf*. There are two constructions, let's see the 'dirty' one first.

If we have open affines $U = \mathrm{Spec}(B) \subseteq X$, $V = \mathrm{Spec}(A) \subseteq Y$, with $f(U) \subseteq V$, then we can define

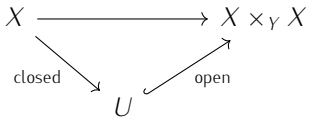
$$\Omega_{X/Y}|_U = \widetilde{\Omega_{B/A}}$$

where $\widetilde{\Omega_{B/A}}$ is the quasicoherent $\mathcal{O}_U$-module attached to the B-module $\Omega_{B/A}$. The map d is induced by

$$\begin{array}{ccc} \mathcal{O}_X(U) & \xrightarrow{\mathrm{d}} & \Omega_{X/Y}(U) \\ \cong\downarrow & & \downarrow\cong \\ B & \xrightarrow[\mathrm{d}]{} & \Omega_{B/A} \end{array}$$

Functoriality means that we can glue to get a quasicoherent sheaf on X. We will omit the details.

For the second construction, consider the diagonal map $\Delta : X \to X \times_Y X$. This map is an immersion, and so it factors as



Let $i : X \hookrightarrow U$ be the closed immersion. Define

$$\Omega_{X/Y} = i^*\left(\mathcal{I}_{X/U}/\mathcal{I}_{X/U}^2\right)$$

where $\mathcal{I}_{X/U}$ is the ideal sheaf. When $X = \mathrm{Spec}(B)$ and $Y = \mathrm{Spec}(A)$, this is just $\widetilde{\Omega_{B/A}}$.

Lecture 3

Note that we also have that $\Omega_{X/Y} = i^*\mathcal{I}_{X/U}$, since

$$i^*\mathcal{I}_{X/U} = \mathcal{O}_X \otimes i^{-1}\mathcal{I}_{X/U} = i^{-1}(\mathcal{O}_U/\mathcal{I}_{X/U}) \otimes i^{-1}\mathcal{I}_{X/U}$$

If $X \to Y$ is separated, then we can take $U = X \times_Y X$.

We will define $\mathrm{d} : \mathcal{O}_X \to \Omega_{X/Y}$ by: if we have open affines $V = \mathrm{Spec}(B) \subseteq X$, $U = \mathrm{Spec}(A) \subseteq Y$, with $f(V) \subseteq U$, then

$$\mathrm{d} : \mathcal{O}_X(V) = B \to \Omega_{B/A} = \Omega_{X/Y}(V)$$

is defined. Moreover, this is compatible with restriction to smaller open affines. The set of open affines $\mathrm{Spec}(B) \subseteq X$ such that the image is contained in an open affine in Y, forms a basis for the topology on X. Thus, this extends to a map of sheaves.

Amusing fact: There exists an immersion $X \hookrightarrow Z$ (i.e. a closed immersion then an open immersion), which cannot be factored as an open immersion, then a closed immersion.

We can rewrite the exact sequences, in the setting of sheaves.

- if we have morphisms of schemes $f : X \to Y$, $g : Y \to S$, then we have

$$f^*\Omega_{Y/S} \longrightarrow \Omega_{X/S} \longrightarrow \Omega_{X/Y} \longrightarrow 0$$

- If $Z \hookrightarrow X$ is a closed immersion, with ideal sheaf $\mathcal{I}$, and $f : X \to Y$ a morphism of sheaves, then we have the exact sequence

$$\mathcal{I}/\mathcal{I}^2 \longrightarrow i^*\Omega_{X/Y} \longrightarrow \Omega_{Z/Y} \longrightarrow 0$$

By construction, $\Omega_{X/Y}$ is quasicoherent.

Proposition 1.10. If $f : X \to Y$ is of finite type, and X and Y are Noetherian, then $\Omega_{X/Y}$ is coherent[a].

[a] i.e. affine locally it is the sheaf associated to a finite module.

Proof. Locally, we can cover X with $\mathrm{Spec}(B)$, such that $f(\mathrm{Spec}(B)) \subseteq \mathrm{Spec}(A)$, and this makes B a finite type A-algebra. That is,

$$B = \frac{A[t_1, \dots, t_m]}{I}$$

where $I = \langle f_1, \dots, f_n \rangle$. The second exact sequence is

$$I/I^2 \longrightarrow \Omega_{A[\mathbf{t}]/A} \otimes_{A[\mathbf{t}]} B \longrightarrow \Omega_{B/A} \longrightarrow 0$$

But note that

$$\Omega_{A[\mathbf{t}]/A} \otimes_{A[\mathbf{t}]} B = \bigoplus_i B\mathrm{d}t_i$$

and so $\Omega_{B/A}$ is a finite B-module, and so $\Omega_{X/Y}$ is coherent. □

For field extensions, Ω tells us whether the extension is separable.

Example 1.11

Let L/K be a finite field extension. Then L/K is separable if and only if $\Omega_{L/K} = 0$. To see this, factor the extension as $K \subseteq K_1 \subseteq L$, with K_1/K separable, and L/K_1 purely inseparable. That is, K_1 is the set of all elements of L which is separable over K.

By the primitive element theorem, $K_1 = K(\alpha) = K[t]/\langle g \rangle$, where $g \in K[t]$ is irreducible, $g(\alpha) = 0$ and $g'(\alpha) \neq 0$. So

$$\Omega_{K_1/K} = \frac{K_1\mathrm{d}\alpha}{K_1 g'(\alpha)\mathrm{d}\alpha}$$

by corollary 1.9. But this is zero as $g'(\alpha) \neq 0$. In this case, transitivity gives us $\Omega_{L/K} = \Omega_{L/K_1}$. If $L = K_1$, i.e. L/K is separable, then $\Omega_{L/K_1} = 0$.

Conversely, if L/K is inseparable, then there exists K_2 with $K_1 \subseteq K_2 \subseteq L$, such that $L = K_2(\beta) = K_2[t]/\langle t^p - b \rangle$, with $p = \mathrm{char}(K) > 0$, $b \in K_2$ is not a p-th power. In this case.

$$\Omega_{L/K_2} = L\mathrm{d}\beta$$

since in this case, $\frac{\mathrm{d}}{\mathrm{d}t}(t^p - b) = 0$. Thus, by transitivity again, $\Omega_{L/K} \neq 0$.

1.2 Tangent and cotangent spaces

Let X be a k-scheme, locally of finite type. That is, X has an open cover by affines $\mathrm{Spec}(A)$, where A is a finite type k-algebra. Let $|X|$ denote the set of closed points of X. If $x \in |X|$, then the residue field at k,

$$\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$$

is a finite extension of k. Define

$$\Omega_{X/k}(x) = \Omega_{X/k,x} \otimes_{\mathcal{O}_{X,x}} \kappa(x)$$

This is a finite dimensional $\kappa(x)$ vector space. This follows from the fact that $\Omega_{X/k}$ is coherent. We call this the *cotangent space of X at x*.

The corresponding *tangent space* is

$$\mathrm{T}_{X,x} = \mathrm{Hom}_{\kappa(x)}(\Omega_{X/k}(x), \kappa(x))$$

Proposition 1.12. If $\kappa(x) = k$, then there exists a canonical isomorphism

$$\Omega_{X/k}(x) \cong \mathfrak{m}_x/\mathfrak{m}_x^2$$

of k-vector spaces.

Proof. In this case,

$$\Omega_{X/k}(x) = \Omega_{\mathcal{O}_{X,x}/k} \otimes_{\mathcal{O}_{X,x}} k$$

The second exact sequence gives us that

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow \Omega_{\mathcal{O}_{X,x}/k} \otimes k \longrightarrow \Omega_{k/k} = 0 \longrightarrow 0$$

It suffices to show that the first map is injective, which in turn, means that it suffices to show that the dual

$$\mathrm{Hom}_{\mathcal{O}_{X,x}/k}(\Omega_{\mathcal{O}_{X,x}/k}, k) \to \mathrm{Hom}_k(\mathfrak{m}_x/\mathfrak{m}_x^2, k)$$

is surjective. The left hand side can be identified with

$$\mathrm{Der}_k(\Omega_{X,x}, k)$$

and so the map is given by

$$\mathrm{D} \mapsto \left(f + \mathfrak{m}_x^2 \mapsto \mathrm{D}(f) \mod \mathfrak{m}\right)$$

Since $\mathrm{D}(\mathfrak{m}_x^2) = 0$ by Leibniz. As $\mathcal{O}_{X,x}/\mathfrak{m}_x = k \hookrightarrow \mathcal{O}_{X,x}$, we must have that

$$\mathcal{O}_{X,x}/\mathfrak{m}^2 = k \oplus \mathfrak{m}_x/\mathfrak{m}_x^2$$

as k-vector spaces. Let $\pi : \mathcal{O}_{X,x}/\mathfrak{m}_x^2 \to \mathfrak{m}_x/\mathfrak{m}_x^2$ be the projection. For $\varphi : \mathfrak{m}_x/\mathfrak{m}_x^2 \to k$, the map $\mathrm{D} = \varphi \circ \pi$ is a derivation which maps to φ.

To show that it is a derivation, recall the ring structure on $k \oplus \mathfrak{m}_x/\mathfrak{m}_x^2$ from before. In particular,

$$(a, b)(a', b') = (aa', ab' + a'b)$$

□